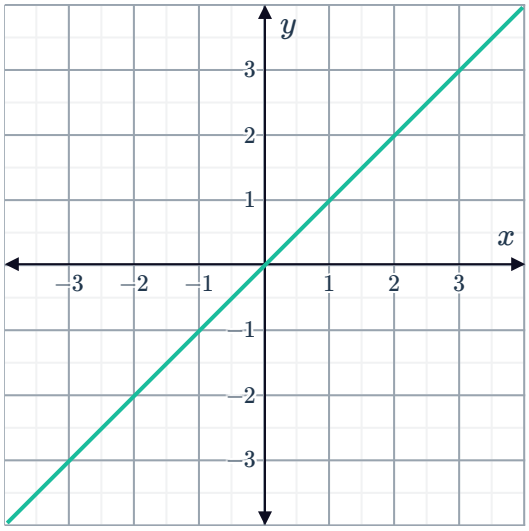
Intercepts

1 For each of the following graphs:

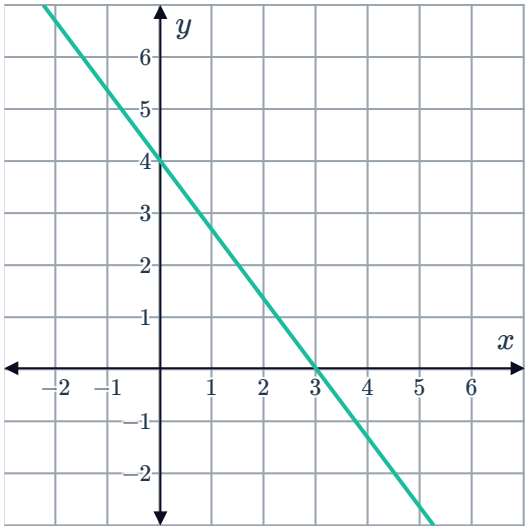
i State the value of the x-intercept.

ii State the value of the y-intercept.

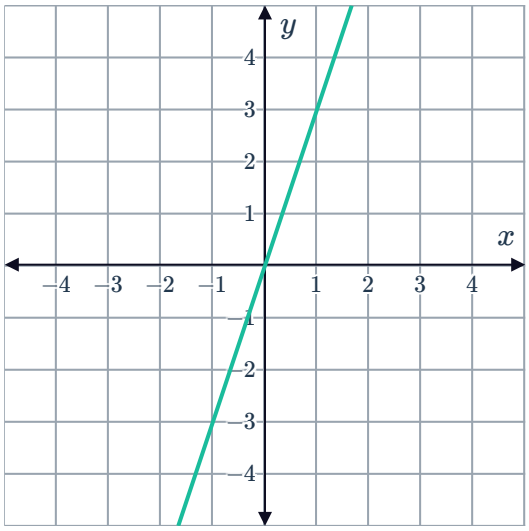
a



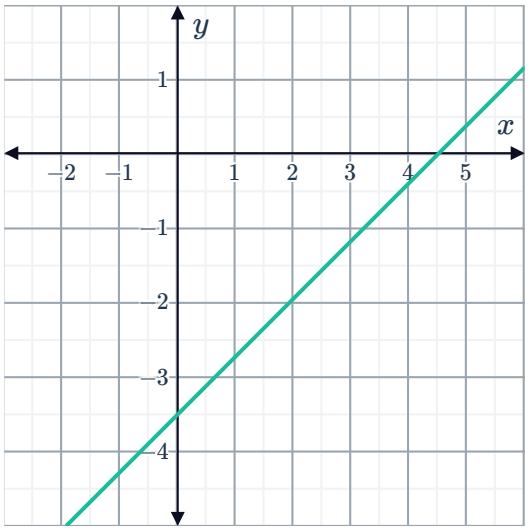
b



c



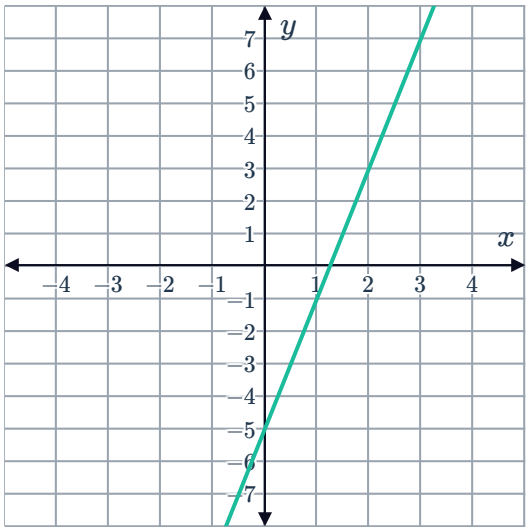
d



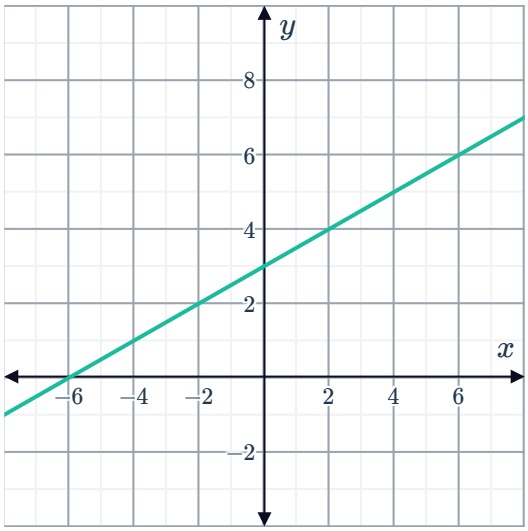
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2 Given each linear equation and its graph, state the coordinates of the y-intercept:

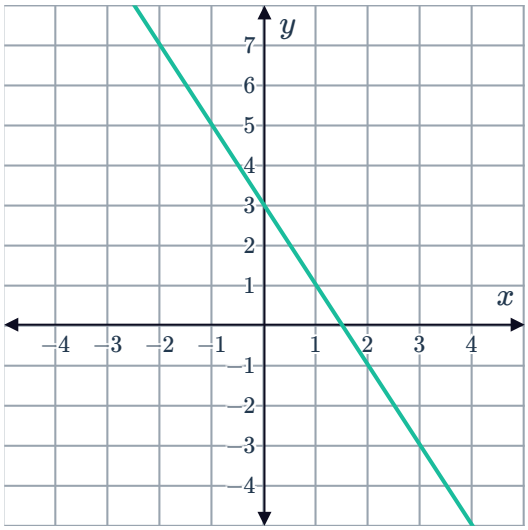
a $y = 4x - 5$



b $y = \frac{x}{2} + 3$



3 Consider the following graph of the line $y = -2x + 3$:



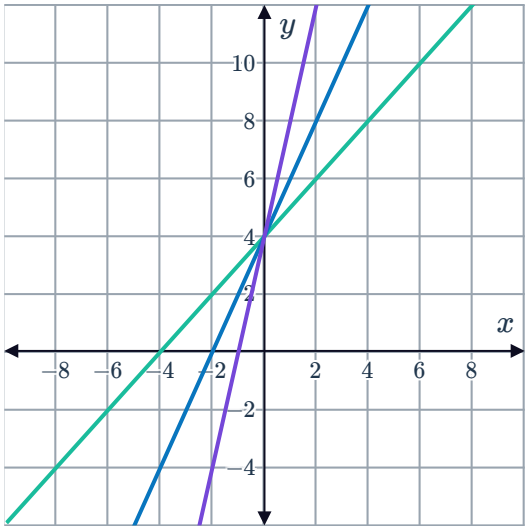
a State the the y-value, when x is 0.

b Explain the relationship between the value of the y-intercept and the equation of the line.

c If the equation of a line is $y = mx + c$, state the value of the y-intercept.

4 Consider the following three linear equations and their corresponding graphs:

$$y = x + 4,\ y = 2x + 4,\ y = 4x + 4$$



a What do all of the equations have in common?

b What do all of the graphs have in common?

c What conclusion can be made about lines that have the form $y = mx + 4$?

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5 Find the value of the y-intercept of the following lines:

a $y = 7x + 3$ **b** $y = 3x - 5$ **c** $y = -8x - 3$ **d** $y = -8x + 4$

e $y = 9x$ **f** $y = 2$ **g** $y = 2x + \frac{2}{3}$ **h** $y = \frac{3x + 8}{5}$

6 Determine whether the following equations represent lines that will cross the y-axis at 2:

a $y = 5x + 2$ **b** $y = 2 - x$ **c** $y = 2x$ **d** $y = 2x - 4$

e $y = x + 2$ **f** $y = x - 2$ **g** $y = 2$ **h** $y = \frac{x + 4}{2}$

7 The x-intercept occurs when $y = 0$. Find the value of the x-intercept for the following lines:

a $y = 2x - 2$ **b** $3x + y = -3$ **c** $y = 4x - 8$ **d** $2y + x = -3$

e $y = 9x$ **f** $2y + 2x = 4$ **g** $3x - 5y = 1$ **h** $x = \frac{3y + 10}{5}$

8 For each of the following equations:

ii Find the coordinates of the y-intercept.

i Find the coordinates of the x-intercept.

iii Use the intercepts to sketch the graph of the line.

a $y = 2x - 4$ **b** $y = -2x + 2$ **c** $y = 3x - 3$ **d** $y = -4x + 8$

Lines through the origin

9 Consider the linear equation $y = 5x$.

a Find the coordinates of the y-intercept. **b** Find the coordinates of the x-intercept.

c Find the value of y when $x = 2$. **d** Hence, sketch the graph of the line.

10 Consider the linear equation $y = -\frac{5x}{4}$.

a Find the coordinates of the y-intercept.

b Find the coordinates of the point on the line where $x = 4$.

c Hence, sketch the graph of the line.

11 If a line has equation $y = mx + c$, explain how you can tell if the line will pass through the origin.

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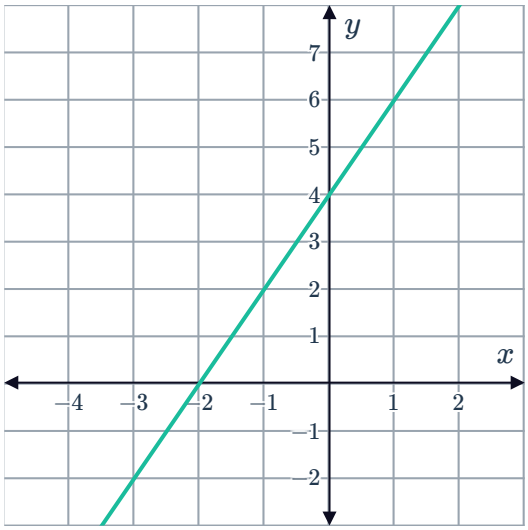
12 Determine whether the following equations represent lines that will pass through the origin:

a $y = 8x - 8$ **b** $y = \frac{x}{8}$ **c** $y = -6x$ **d** $y = 8x$

e $y = \frac{x}{8}$ **f** $y = 0$ **g** $y = -x$ **h** $y = -6x - 6$

Gradient

13 Consider the line graph shown:



a State the y-value when $x = 0$.

b State the y-value when $x = 1$.

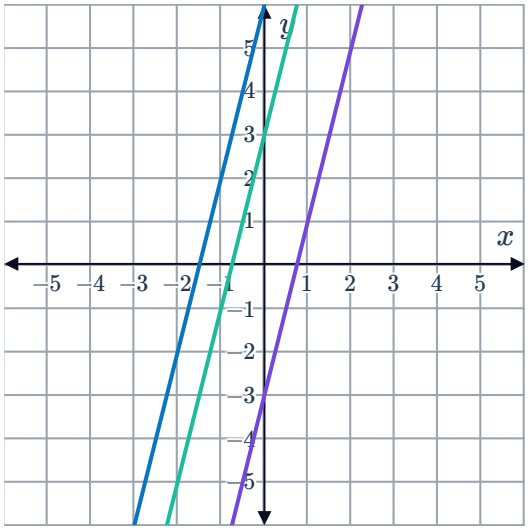
c When the x-value increases by 1, by how much does the y-value change?

d Hence state the gradient of the line, m.

e The equation of this line is $y = 2x + 4$. Explain how to find the gradient from the equation of the line.

14 Consider the following three linear equations and their corresponding graphs:

$$y = 4x + 3,\ y = 4x + 6,\ y = 4x - 3$$



a What do all of the equations have in common?

b What do all of the graphs have in common?

c What conclusion can be made about lines that have the form $y = 4x + c$?

15 Find the gradient, m, of the following linear equations:

a $y = 9x + 3$ **b** $y = -7x + 5$ **c** $y = \frac{5x}{4} + 2$ **d** $y = -x + 5$

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16 From the following list of equations, select the lines that have the same gradient:

- $y = 2 + 7x$
- $y = \frac{x}{7} + 2$
- $y = 5 - 7x$
- $y = 7x$
- $y = 5x + 7$
- $y = 7x - 2$

17 For each linear equation:

i Find the value, m, of the gradient.

ii Find the value, c, of the y-intercept.

a $y = 2x + 9$ **b** $y = 5x - 6$ **c** $y = -5x + 8$ **d** $y = -4x - 2$

e $2y = 8x - 1$ **f** $3y = -6x - 2$ **g** $y - 5x = 4$ **h** $2y - 3x = 6$

18 Given the values of m and c, write the equation of the line:

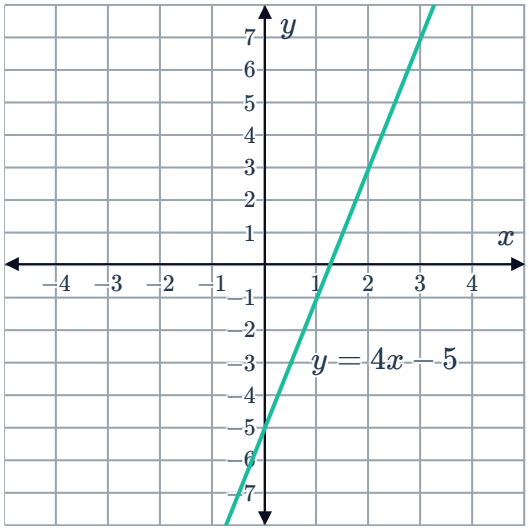
a $m = 2, c = 5$ **b** $m = -3, c = 2$ **c** $m = -2, c = -1$ **d** $m = 4, c = 0$

e $m = 0, c = -7$ **f** $m = 0, c = 4$ **g** $m = \frac{1}{2}, c = -2$ **h** $m = -\frac{3}{4}, c = \frac{1}{2}$

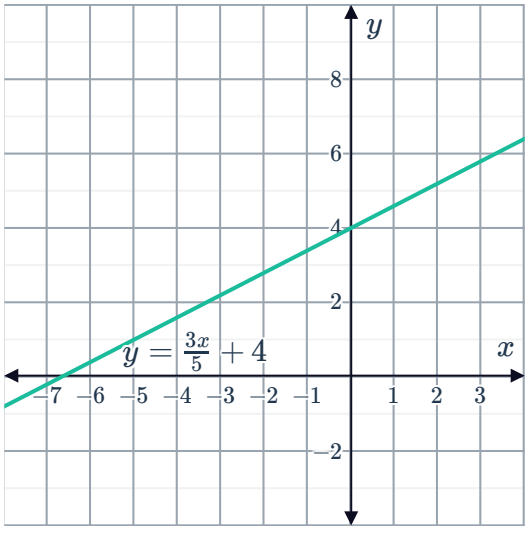
Approximate solutions of equations

19 Approximate the solutions of the following equations using the graphs provided. Round your answers to one decimal place.

a $4x - 5 = 0$

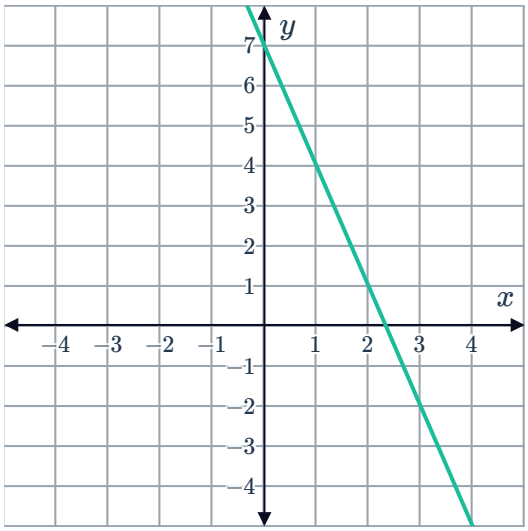


b $\frac{3x}{5} + 4 = 0$



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20 Consider the following graph of the line $y = -3x + 7$:



a Estimate the x-intercept to one decimal place.

b Hence, estimate the solution to the equation $-3x + 7 = 0$ to one decimal place.

c Estimate the x-coordinate of the point on the line where $y = 6$.

d Hence, estimate the solution to the equation $-3x + 7 = 6$ to one decimal place.

21 Estimate the solutions of the following equations by first drawing the appropriate graph. Round your answers to one decimal place.

a $5x - 11 = 0$

b $-4x + 9 = 0$

c $6x - 9 = 0$

d $\frac{9x}{5} - 1 = 0$

e $-\frac{7x}{2} + 10 = 0$

f $\frac{3x}{8} - 5 = 0$

g $4x + \frac{1}{2} = 0$

h $\frac{2x}{3} + 7 = 0$

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